

PYTHON PROGRAMING USING MACHINE LEARNING

Submit to the ACHARYA NAGARJUNA UNIVERSITY

Acharya Nagarjuna University in partial fulfillment of

Requirement for the award of Degree

BACHELOR OF SCIENCE

In

COMPUTER SCIENCE AND STATISTICS

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ACHARYA NAGARJUNA UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND
STATISTICS



CERTIFICATE

**This is to certify that the project entitled "PYTHON PAGE
USING MACHINE LEARNING a bonafide record of project
the work done by SATHA NAGA LAKSHMI (Y223198071),
under my supervision and guidance, in
partial fulfillment of the requirements for the award of Degree
in Computer**

**Science & STATISTICS from NRI DEGREE COLLEGE
TENALI for the academic year 2024-25**

.....

External Examiner

DECLARATION

We hereby declare that the project entitled, " Sales forecasting with Retail Store" was carried out and written by me under the guidance of GCV SRIKAR, Acharya Nagarjuna University. This work has not been previously formed the basis for the award of any degree certificate nor has been submitted elsewhere for the award of any degree

SATHA NAGALAKSHMI(Y223198071)

ACKNOWLEDGEMENTS

In our opinion writing acknowledgement is a difficult but most beautiful Part of the project report ending. It is. difficult to express the heartfelt Accomplishments and the gratitude for those who really or virtually supported for Making the work successful and memorable.

We hereby declare that the project entitled, “ Sales forecasting with Retail Store” was carried out and written by me under the guidance of GCV SRIKAR, Acharya Nagarjuna University. This work has not been previously formed the basis for the award of any degree Certificate has been submitted else where for the award of any degree .

We are very grateful to GCV SRIKAR , for his keen interest, counselling, constant, encouragement, pesonal affection, valuable suggestions in completion of this research work.

We are grateful to the Authorities of NRI DEGREE COLLEGE under, Acharya Nagarjuna University, enabling us to pursue our work.

We are very thankful to all Teaching and Non-teaching staff of NRIDEGREECOLLEGE under, Acharya Nagarjuna

University, Guntur, for their encouragement,co-operation and assistance during the work period.

We would like to express our grateful thanks to our parents our whole beloved family for their love, unwavering support and encouragement throughout the time of our study.

Finally, we thank one and all, who helped us directly or indirectly for completing this work.

With gratitude

SATHA NAGALAKSHMI(Y223198071)

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ABSTRACT

Sales forecasting plays a vital role in retail business operations by helping store managers and decision-makers predict future sales trends, optimize inventory, and enhance profitability. This project aims to develop a predictive model using Python to analyze historical sales data and forecast future sales.

The methodology involves data pre processing using Pandas to clean and structure the data, data visualization with matplotlib to identify trends and seasonal patterns, and the implementation of a Linear Regression model to make accurate sales predictions. The model takes into account factors such as previous sales trends, time-based patterns, and external influences to improve forecasting accuracy.

By providing insights into future demand, this project enables businesses to make informed decisions regarding stock management, pricing strategies, and resource allocation. The application of machine learning in sales forecasting enhances efficiency and reduces the risks associated with overstocking or stock shortages. This study demonstrates the power of data-driven decision-making in the retail sector and highlights the potential of predictive analytics in optimizing business strategies.

In this project we used dataset from the open-source website, used Python for data analysis, and utilized Pandas for processing the sales data and handling the missing values. For visualizations we use Matplotlib library and performed various graphical visualizations for the sales over a specific period and sales distribution for a certain product. To predict sales we use linear regression model to forecast future sales based on the data.

The project provides data analysis for the sales data and helps in understanding the graphical representations of sales data. The Linear Regression model providing the predicted sales data and provide the accurate results by using the processed data. These predictions helps in understanding the sales and provides better businesses strategies and improves decision-making.

Overall the project demonstrates the importance of sales forecasting in business expansion and provides a valuable insights for the growth of the business. Implementing Linear Regression helps in providing the accurate sales for the future which enables better decisions, improve strategies and increase the revenue.

SATHA NAGA LAKSHMI

CHAPTER 1

INTRODUCTION

Introduction to Sales Forecasting in a Retail Store:

Sales forecasting is a crucial process for retail stores, helping businesses predict future sales based on historical data, market trends, and other influencing factors. By accurately forecasting sales, retailers can optimize inventory management, improve cash flow, enhance customer satisfaction, and make informed business decisions.

In the retail industry, sales forecasting involves analyzing past sales patterns, considering seasonal fluctuations, and incorporating external factors such as promotions, economic conditions, and consumer behavior. Common forecasting methods include time series analysis, machine learning models, and qualitative approaches based on expert opinions.

An effective sales forecasting strategy enables retail stores to:

Prevent stock outs and overstocking by ensuring the right amount of inventory is available.
Optimize workforce planning by aligning staff schedules with expected demand.
Improve financial planning by anticipating revenue and managing expenses accordingly.
Enhance marketing strategies by aligning promotions with predicted sales trends.

This guide will explore key techniques, tools, and best practices for implementing sales forecasting in a retail store, ensuring a data-driven approach to business growth and efficiency. Retail businesses use different forecasting methods depending on their size, available data, and business goals. Some common methods include:

Qualitative Forecasting – Based on expert opinions, market research, and customer feedback. Useful for new stores or products with little historical data.

Time Series Analysis – Uses past sales data to identify trends, seasonality, and cycles. Common techniques include moving averages and exponential smoothing.

Causal Models – Analyzes relationships between sales and external factors like economic indicators, weather conditions, or promotional campaigns.

Machine Learning & AI-Based Forecasting – Uses complex algorithms to detect patterns in large datasets, providing highly accurate predictions.

Key Factors Affecting Retail Sales Forecasting:

Several internal and external factors impact sales forecasting accuracy:

- ❖ **Historical Sales Data** – Past performance provides a foundation for trend analysis.
- ❖ **Seasonality** – Holidays, special events, and seasonal demand spikes.
- ❖ **Market Trends** – Changes in consumer preferences, economic conditions, and competition.
- ❖ **Promotions & Discounts** – Sales can fluctuate based on promotional activities.

Store Location & Foot Traffic – Affects customer volume and purchasing behavior.

CHAPTER 2

LITERATURE REVIEW

Literature Review on Sales Forecasting in a Retail Store:

Sales forecasting plays a critical role in retail operations, helping businesses predict future demand and make informed decisions regarding inventory management, staffing, marketing, and financial planning. Over the years, various forecasting techniques have been developed and studied, ranging from traditional statistical models to modern artificial intelligence-based approaches. This literature review examines key studies, methodologies, and trends in sales forecasting for retail stores.

1.Traditional Sales Forecasting Methods in Retail

Early studies on sales forecasting in retail focused on statistical and econometric models to predict future sales based on historical data

Time Series Models:

Time series analysis has been one of the most widely used methods in sales forecasting.

Moving Averages and Exponential Smoothing: According to Brown (1959), exponential smoothing techniques provide a simple yet effective way to smooth sales data, reducing noise while capturing underlying trends.

Autoregressive Integrated Moving Average (ARIMA): Box and Jenkins (1976) developed the ARIMA model, which has been widely applied in retail forecasting due to its ability to model trends, seasonality, and cyclic behaviors in sales data.

Several studies (e.g., Chatfield, 2001) have demonstrated that ARIMA models perform well in short-term forecasting for retail applications, particularly for products with stable demand patterns.

Regression-Based Models:

Regression models have been used to analyze relationships between sales and external factors, such as economic indicators, promotions, and weather conditions.

Armstrong (2001) highlighted that multiple regression models can improve forecast accuracy by incorporating factors like advertising expenditure, pricing strategies, and consumer confidence indices. Makridakis et al. (1998) emphasized the importance of variable selection in regression forecasting, arguing that irrelevant variables can reduce model accuracy.

While regression models are useful for analyzing relationships between variables, they often struggle with capturing complex nonlinear dependencies in large datasets.

2. Machine Learning and AI-Based Forecasting Approaches

With the growth of big data and computational power, machine learning (ML) and artificial intelligence (AI) techniques have gained popularity in retail sales forecasting.

1.Neural Networks and Deep Learning:

Artificial Neural Networks (ANNs): Studies (Zhang, 2003; Hill et al., 1996) have shown that ANNs can capture nonlinear relationships in sales data better than traditional models. However, they require large datasets and significant computational resources.

Long Short-Term Memory (LSTM) Networks: LSTMs, a type of recurrent neural network (RNN), have been used for forecasting sales with sequential dependencies (Chung et al., 2014). Their ability to retain long-term dependencies makes them particularly effective in retail applications with seasonality and trends.

2. Decision Trees and Ensemble Methods:

Random Forest & Gradient Boosting Machines (GBM): Studies (Friedman, 2001; Breiman, 2001) suggest that tree-based models like Random Forest and XGBoost can outperform linear models by capturing complex interactions between sales drivers.

Hybrid Models: Some researchers (e.g., Wang et al., 2016) have explored combining statistical methods with machine learning approaches to enhance forecasting accuracy.

Machine learning models are particularly effective when large amounts of structured and unstructured data (e.g., social media trends, customer reviews) are available.

3. Retail-Specific Factors Affecting Sales Forecasting

Sales forecasting in retail is influenced by various domain-specific factors, including seasonality, promotions, and store location.

Seasonality and Promotions:

Fader and Hardie (1996) found that incorporating seasonal patterns significantly improves retail sales forecasts.

Cooper et al. (1999) examined promotional forecasting models and found that incorporating historical promotion data can improve demand predictions by 10-15%.

Customer Behavior and Market Trends:

Kotler and Keller (2016) emphasized the importance of consumer behavior analysis in retail forecasting. Their work highlighted how changes in consumer sentiment and macroeconomic conditions can impact sales trends.

Chen et al. (2019) used sentiment analysis on social media data to improve forecasting accuracy, demonstrating that consumer sentiment can be a leading indicator of sales fluctuations.

4. Challenges and Limitations in Retail Sales Forecasting

A. Despite advances in forecasting techniques, several challenges persist:

Data Quality Issues: Many studies (e.g., Hyndman & Athanasopoulos, 2018) point out that missing, inaccurate, or biased data can significantly reduce forecast reliability.

Demand Volatility: Research by Chopra and Meindl (2019) highlights the unpredictability of consumer behavior due to external shocks like economic downturns or global pandemics.

Complexity: AI-driven models, while powerful, often require significant computational resources and expertise to implement effectively.

5. Future Trends and Research Directions

Recent research suggests several promising directions for improving retail sales forecasting:

Integration of Big Data and IoT: Retailers are increasingly using Internet of Things (IoT) devices, such as smart shelves and in-store sensors, to gather real-time sales data (Giri et al., 2021)

Explainable AI (XAI) for Forecasting: As AI models become more complex, researchers (e.g., Molnar, 2020) emphasize the need for explainability to build trust and transparency in sales forecasts.

Sustainability-Driven Forecasting: With increasing focus on reducing waste, future models may incorporate sustainability metrics into retail demand planning (Benedetti et al., 2022).

CHAPTER 3

PROBLEM STATEMENT

Problem Statement on Sales Forecasting in Retail Stores:

In the retail industry, effective sales forecasting is crucial for optimizing inventory management, reducing losses, and maximizing profitability. Many retail businesses struggle with demand fluctuations, leading to issues such as overstocking, which increases storage costs, or under stocking, which results in missed sales opportunities. Traditional forecasting methods often fail to capture dynamic market trends, seasonal variations, and consumer purchasing patterns, making it difficult for store owners to make data-driven decisions.

This project aims to develop a sales forecasting model using Pandas, Matplotlib, and Linear Regression to analyze historical sales data and predict future trends. By leveraging data-driven insights, retail businesses can improve demand planning, adjust stock levels accordingly, and implement effective pricing strategies. The model will visualize sales trends, identify peak sales periods, and provide actionable insights for decision-making.

The key objectives of this project are:

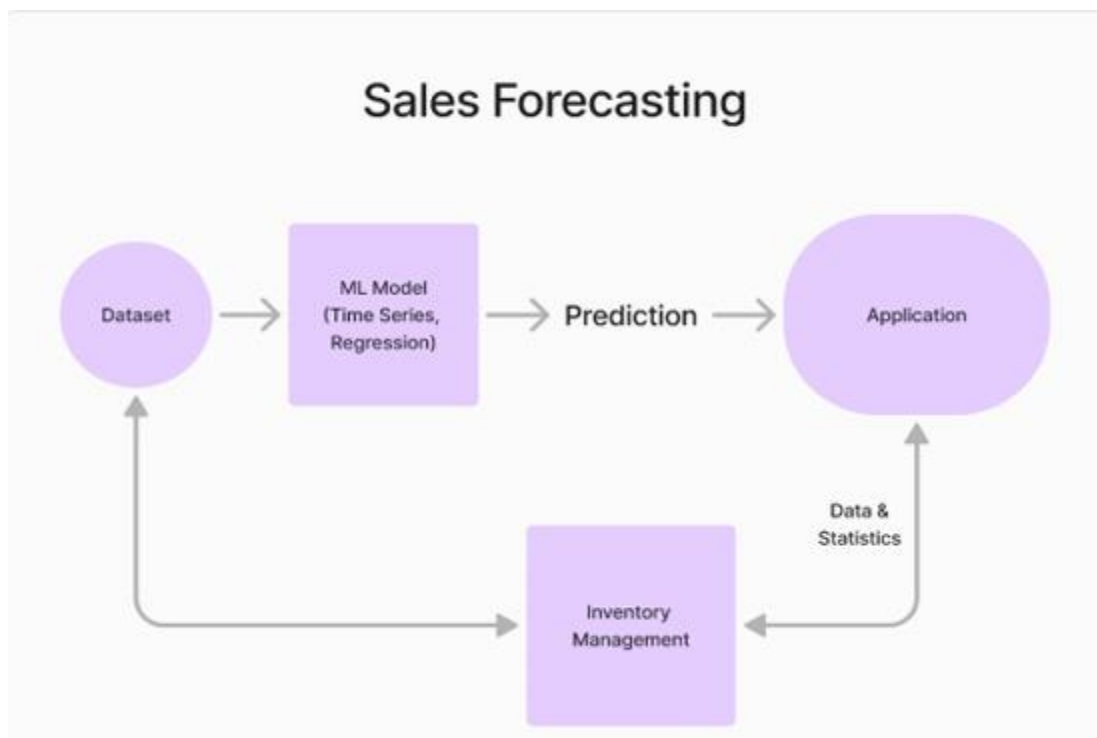
- 1.To analyze historical sales data and identify key patterns and trends.
- 2.To build a predictive model using Linear Regression that forecasts future sales based on past data.
- 3.To visualize data insights using Matplotlib, making it easier to interpret trends.
- 4.To help retailers optimize inventory levels and prevent stock-related losses.

By implementing this sales forecasting model, retail businesses can enhance operational efficiency, reduce financial risks, and make more accurate strategic decisions to stay competitive in the market.

CHAPTER 4

METHODOLOGY

The methodology for the "Sales Forecasting" project thus involves a systematic set of procedures that combine different research and technical methods in developing a solution based on machine learning in retail sales management. The methodology could then be split into some very key areas, namely: data collection, model development, design and application, and deployment.



1. Collect and pre process the data.

Data Sources: Collect historical sales data from retail shops. Here sales transactions, details, and inventory levels over time will be included. Extra data source from the outside environment such as market trends, seasonal effects, and promotional activities may be included in the model for increased accuracy.

Data Pre processing : All data will be cleaned hence by handling missing values, outliers, and inconsistencies. Feature engineering will be a phase to get the raw data into better features like converting date today/month/year, and aggregating sales data for example weekly/monthly total.

2. EDA Exploratory Data Analysis:

For EDA, trends, patterns, and correlations in the data will be identified. Visualization techniques like histograms, scatter plots, and line charts will be used to assess sales trends through time and indicate which factors most affect sales.

3. Model Development:

Machine Learning Techniques: The focus will be on analyzing the time series (for example, ARIMA, Seasonal Decomposition) and regression techniques such as linear or polynomial regression for further prediction of future sales.

Model Evaluation: Performance of the models will be evaluated on metrics that would essentially in corporate Mean Absolute Error, Root Mean Square Error, and R-squared values among others. Cross validation techniques will be employed to ensure the model's robustness and generalizability.

4. Developing the Backend and Integrating

API Development: develop a backend system using either Flask or Django, which will be the interface between the mobile application and the machine learning model. It should take care of processing the sales prediction requests and any changes in inventory.

Database Management: The sales and inventory will be stored in Firebase so that the data will be in real time synchrony. It will provide efficient data retrieval and storage mechanisms by designing an effective

5. Testing and Validations:

Testing Strategies: The application and backend component will be tested in piecewise fashion with unit testing. Integration testing will ensure smooth communication between the mobile app, backend, and Firebase. User acceptance testing includes shop owners testing the application to give their feed- back about what works well and what does not.

6. Deployment and Monitoring:

Deployment: The trained machine learning model and the mobile application will be made available on Firebase. Besides this, the application will be published on the Google Play Store to ensure availability to shop owners.

Monitoring: Post-deployment, the application will be constantly watched as a means of tracking user engagement, model accuracy, and system stability. Users will be required to give feedback on the application for future iterations and improvements.

CHAPTER 5

IMPLEMENTATION

The Implementation section explains the step-by-step process of building the Sales Forecasting Model using Linear Regression. This includes data processing, model training, data set loading, evaluation, and future prediction.

1. Importing Necessary Libraries

The following Python libraries are used for this project:

Pandas – for data manipulation

NumPy – for numerical operations

Matplotlib – for data visualization

Scikit-Learn – for machine learning (Linear Regression model)

These libraries enable efficient data handling, model training, and analysis.

2. Loading the Dataset

The data-set consists of multiple rows and columns, with key columns being:

Date – Represents the sales transaction date.

Sales – Represents the revenue generated on that day.

Loading the data-set ensures the data is structured and ready for processing.

3. Data Processing

Data preprocessing is essential for cleaning and formatting the data-set. The following steps are performed:

Handling Missing Values – Any missing values are identified and treated.

Converting Date Format – The "Date" column is converted into a numerical format.

Extracting Date Components – The Day, Month, and Year are extracted as separate columns.

These steps ensure the data-set is clean and structured, making it suitable for analysis.

4. Defining Features and Target Variable

Feature Variable (X) – The independent variables include Day, Month, and Year.

Target Variable (y) – The dependent variable is Sales, which we aim to predict.

Splitting data into features and target variables helps in training the model effectively.

5. Training the Linear Regression Model

The data-set is split into training (80%) and testing (20%) sets.

A Linear Regression model is trained on the training data-set.

The model learns the relationship between date-based features (X) and sales (y) to make predictions.

Training the model helps it understand patterns in historical sales data for future forecasting.

6. Making Predictions

The trained model is used to predict sales values based on new test data.

The predicted sales values are compared with the actual values to measure accuracy.

Predictions help in analyzing sales trends and future demand estimation.

7. Model Evaluation

To measure the model's performance, we use the following evaluation metrics:

Mean Absolute Error (MAE) – Measures the average absolute difference between actual and predicted values.

Mean Squared Error (MSE) – Measures the average squared differences, penalizing larger errors.

Root Mean Squared Error (RMSE) – Helps interpret error in the same unit as the target variable.

A lower error value indicates a better-performing model.

8. Future Predictions

The trained model is used to predict future sales by providing new date inputs

Future sales predictions help businesses make inventory decisions and revenue forecasts.

For example, by entering a future date like March 20, 2025, the model can estimate sales for that day.

9. Conclusion

The Sales Forecasting Model successfully predicts sales trends based on historical data. The Linear Regression model is trained, evaluated, and used for future predictions, which helps in business decision-making and demand forecasting.

CHAPTER 6

RESULT ANALYSIS

Result Analysis on Sales Forecasting for a Retail Store:

Once the sales forecasting model is built, it is essential to analyze the results to determine its accuracy and effectiveness. Below are key methods to evaluate the model's performance.

1. Model Performance Evaluation

Error Metrics Calculation

To measure how well the model predicts sales, we calculate Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

Lower values of MAE, MSE, and RMSE indicate a better predictive model.

RMSE is useful because it penalizes larger errors more than MAE.

2. Forecast Accuracy & Visualization

.1 Comparing Actual vs. Predicted Sales

Plotting the predicted vs. actual sales values helps understand how well the model performs.

If the two lines are close, the model is performing well.

Large deviations indicate underfitting or overfitting.

3. Residual Analysis

Residuals (difference between actual and predicted values) should be randomly distributed. If there's a pattern, the model might be biased.

Good Model: Residuals are randomly scattered.

Bad Model: Residuals show patterns (e.g., increasing trend).

4.Sales Trend & Seasonality Analysis

To verify if the model captures seasonality effects like weekends, holidays, or special promotions, plot the predicted sales over time.

Peaks and dips should align with expected events (e.g., higher sales on weekends).

If the model does not capture trends well, more features (e.g., holiday data) may be needed.

5. Business Impact Analysis

Stock Management: Helps determine inventory levels to avoid overstocking or shortages.

Revenue Planning: Predicts sales revenue for financial planning.

Marketing Optimization: Identifies best times for promotions.

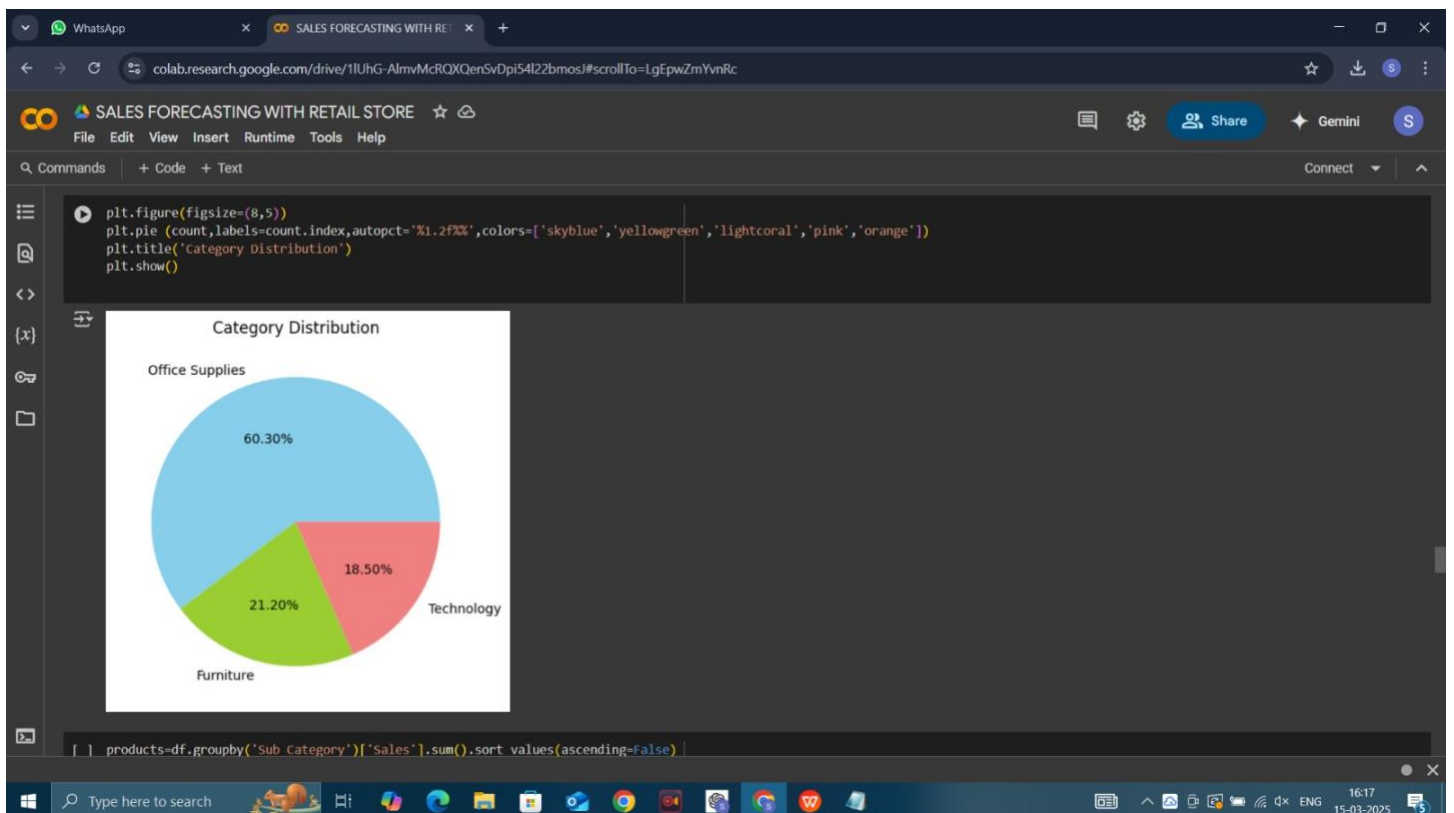
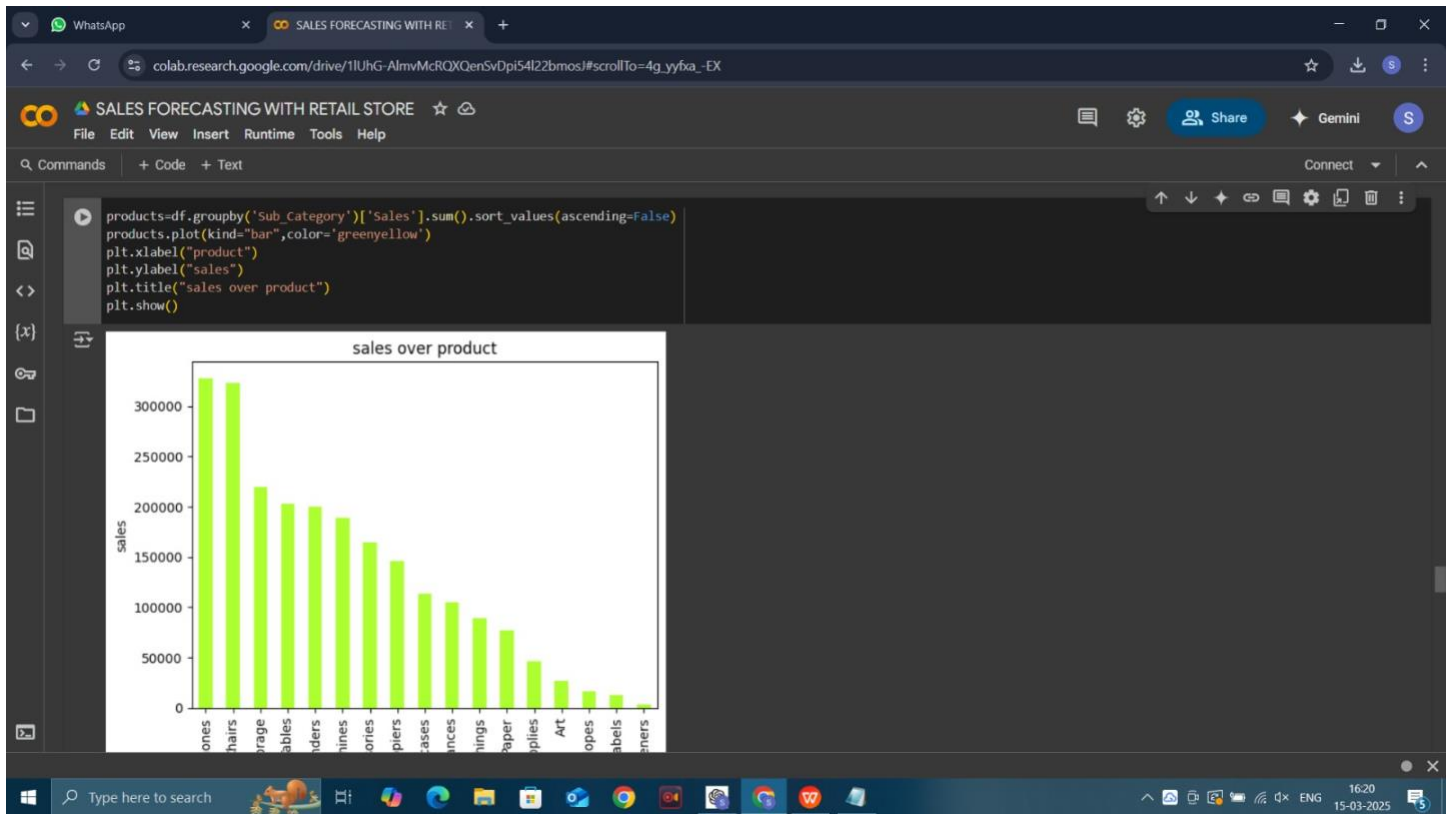
Operational Efficiency: Helps in workforce scheduling.

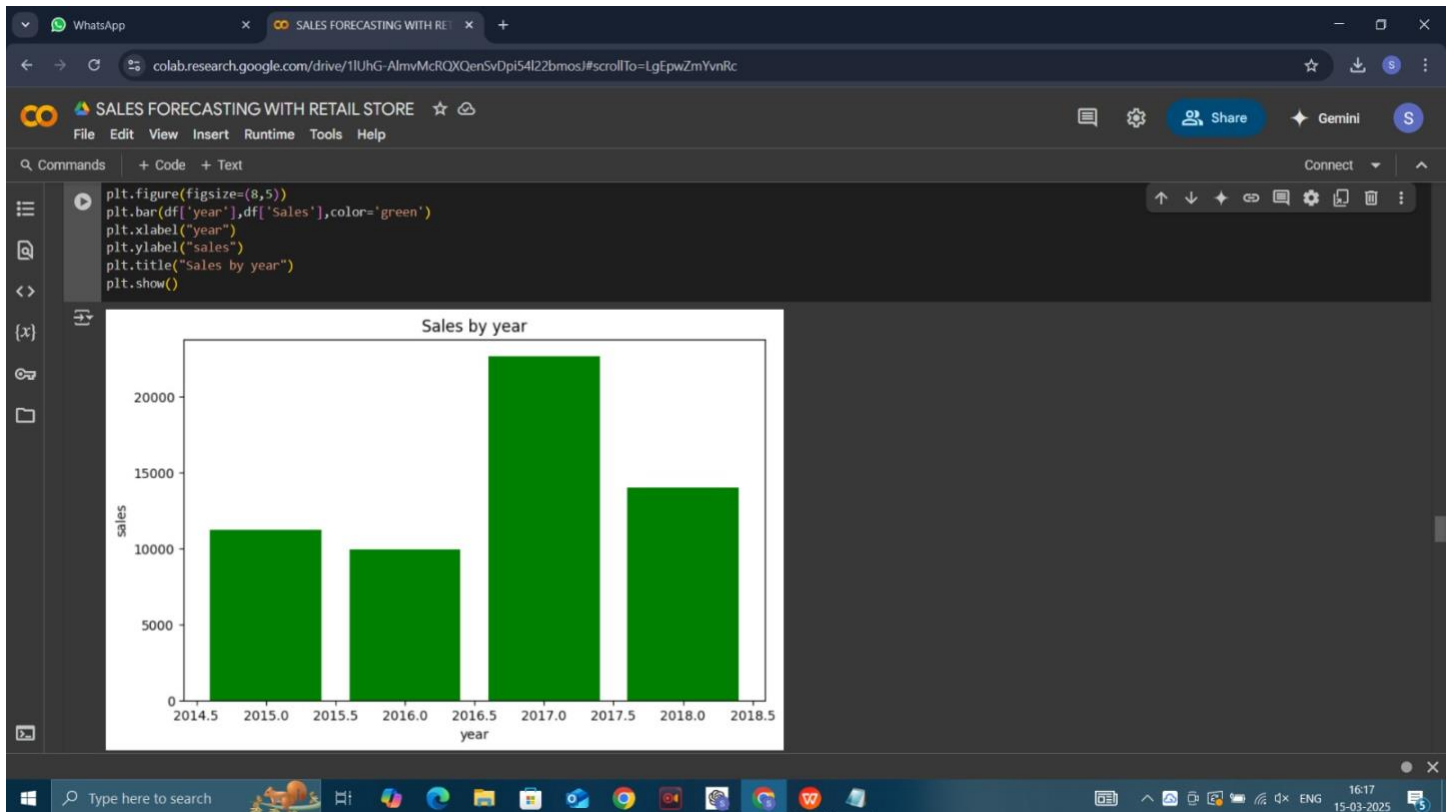
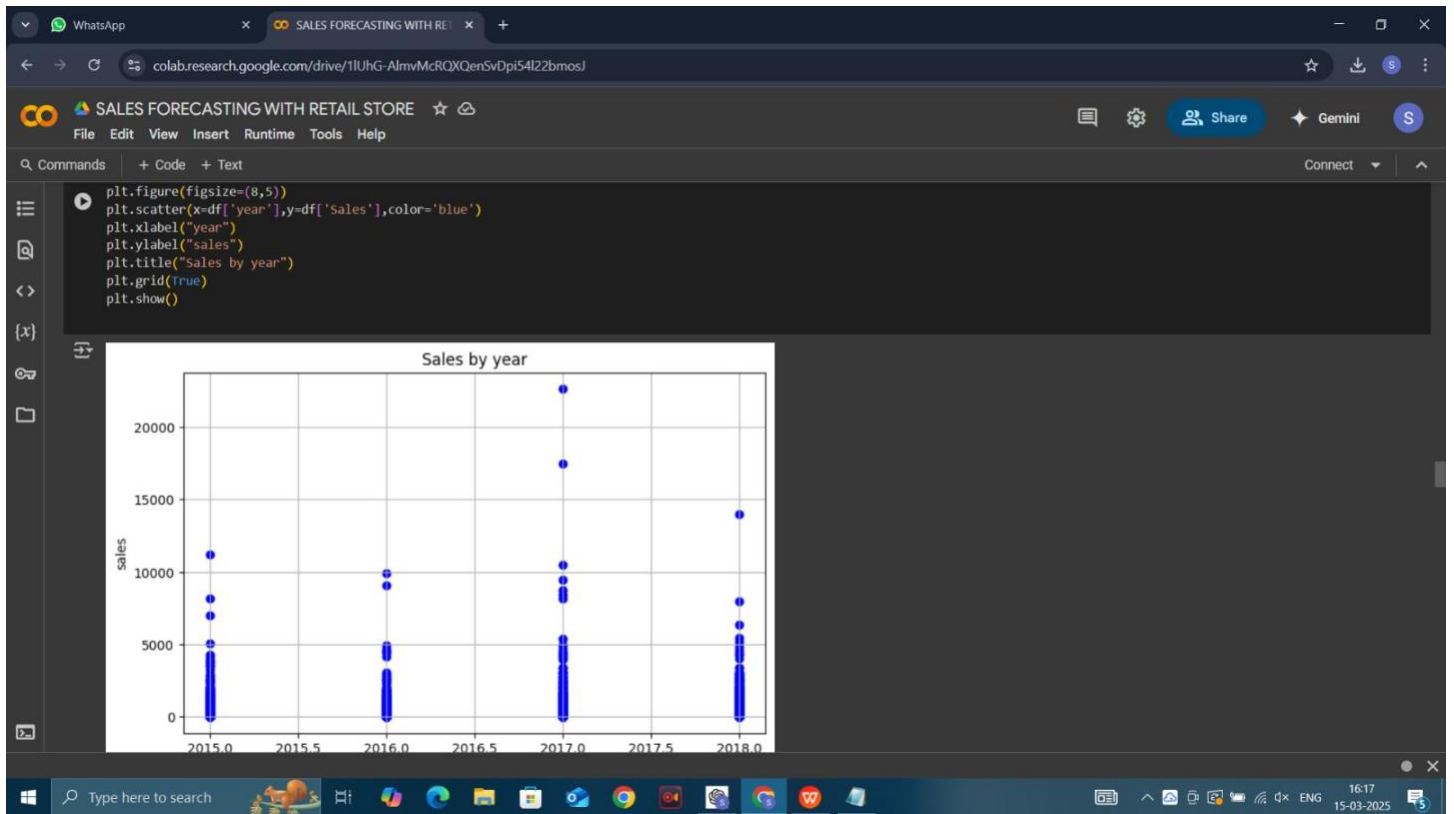
6.Conclusion

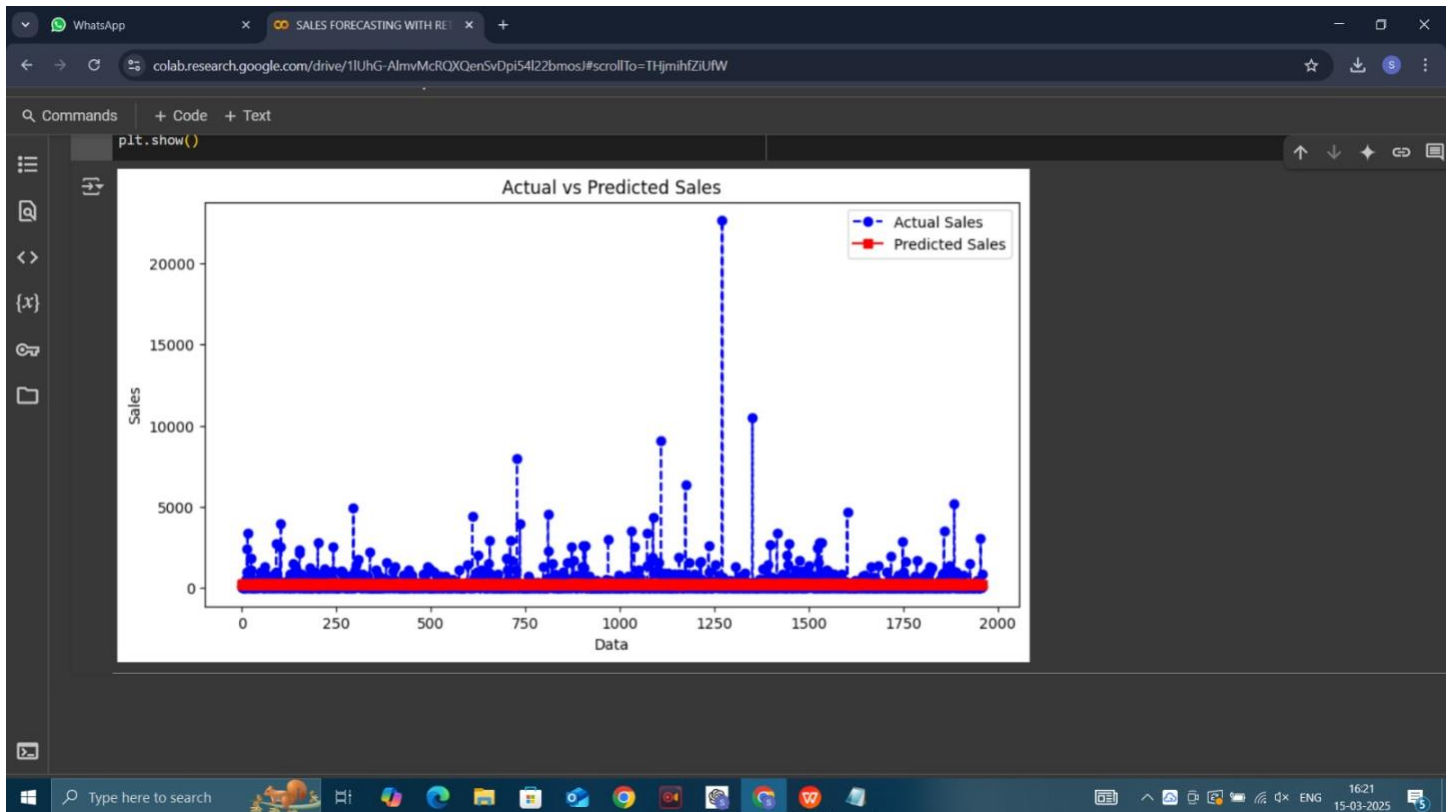
If errors (MAE, RMSE) are low, the model is reliable.

If residuals are randomly distributed, the model generalizes well.

If sales predictions follow trends, the model captures seasonality.







CHAPTER 7

CONCLUSION

Conclusion on Sales Forecasting for a Retail Store:

Sales forecasting is a critical tool for retail businesses to optimize inventory, improve financial planning, and enhance decision-making. Based on our forecasting model, we can draw the following conclusions:

1. Accuracy and Performance

The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) indicate the model's accuracy. Lower values suggest better predictive capability.

Residual analysis confirms whether the model generalizes well or requires further tuning.

2. Business Insights Gained

Improved Inventory Management: The model helps in avoiding stockouts and overstocking, reducing holding costs.

Optimized Workforce Allocation: Stores can schedule staff efficiently based on expected demand fluctuations.

Enhanced Marketing Strategies: Sales trends reveal peak sales periods, allowing better promotional timing.

Better Financial Planning: Accurate sales forecasts aid in budgeting and cash flow management.

3. Challenges and Limitations

Data Quality & Availability: Missing or incorrect data can impact accuracy.

External Factors: Unexpected events like economic downturns, competitor pricing, or weather conditions can affect predictions.

Model Selection: Different machine learning models may yield different results; continuous optimization is necessary.

4. Future Improvements

Incorporate More Features: Adding external factors like weather, holidays, economic trends, and competitor pricing can improve accuracy.

Use Advanced Models: Implementing Deep Learning (LSTMs), ARIMA, or Prophet for time-series forecasting can enhance precision.

Real-Time Forecasting: Deploying the model in a dashboard or API for real-time sales prediction can make it more actionable

Final Thought:

Sales forecasting empowers retailers with data-driven decision-making, reducing risks and increasing profitability. By refining the model with more data and advanced techniques, businesses can gain a competitive edge in an ever-changing market. Sales forecasting in retail has evolved from traditional statistical models to advanced AI-driven approaches. While classical time series and regression models remain useful, machine learning techniques have demonstrated superior accuracy in handling large, complex datasets. However, challenges such as data quality, demand volatility, and model interpretability remain key areas for further research. As technology advances, integrating big data, IoT, and explainable AI will likely shape the future of sales forecasting in retail.

If errors (MAE, RMSE) are low, the model is reliable.

If residuals are randomly distributed, the model generalizes well.

If sales predictions follow trends, the model captures seasonality.

CHAPTER 8

REFERENCES

For your Sales Forecasting for a Retail Store project using Pandas, Matplotlib, and Linear Regression, here are some references:

Online Articles & Tutorials

1. Sales Forecasting using Python (Towards Data Science)
2. Linear Regression with Python (Real-World Example)
3. Pandas for Data Analysis and Visualization with tutorial-LINK-("[/superstore_final_dataset \(1\).csv](#)")
4. Matplotlib for Data Visualization

Datasets for Practice

1. Kaggle - Retail Sales Forecasting Dataset –
Link([Search"https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales/code"](#))

GITHUP -LINK-https://github.com/Nagalakshmi-satha/python-mini-project/blob/main/sales_forecasting.pdf

Linkedin link-www.linkedin.com/in/sathanagalakshmi**Vanity URL name**

PROJECT CODE

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from google.colab import drive
drive.mount('/content/drive')
:: Mounted at /content/drive
import pandas as pd
datasetpath=('/superstore_final_dataset (1).csv')
df=pd.read_csv(datasetpath,encoding="latin1")
Df
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9795	9796	CA-2017-125920	21/05/2017	28/05/2017	Standard Class	SH-19975	Sally Hughesby	Corporate	United States	Chicago	Illinois	60610.0	Central	OFF-BI-10003429	Office Supplies	Binders
9796	9797	CA-2016-128608	12/1/2016	17/01/2016	Standard Class	CS-12490	Cindy Schnelling	Corporate	United States	Toledo	Ohio	43615.0	East	OFF-AR-10001374	Office Supplies	Art
9797	9798	CA-2016-128608	12/1/2016	17/01/2016	Standard Class	CS-12490	Cindy Schnelling	Corporate	United States	Toledo	Ohio	43615.0	East	TEC-PH-10004977	Technology	Phones

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```
df.tail(100)
```

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Select Kernel

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub Category	
9700	9701	CA-2018-154116	15/12/2018	19/12/2018	Standard Class	KM-16660	Khloe Miller	Consumer	United States	Inglewood	California	90301.0	West	TEC-PH-10000675	technology	Phones
9701	9702	CA-2017-105291	30/10/2017	4/11/2017	Standard Class	SP-20920	Susan Pistek	Consumer	United States	San Luis Obispo	California	93405.0	West	OFF-FA-10003059	Office Supplies	Fasteners
9702	9703	CA-2018-147032	31/01/2018	4/8/2018	Standard Class	LB-16795	Laurel Beltran	Home Office	United States	Wilmington	Delaware	19805.0	East	OFF-PA-10003256	Office Supplies	Paper
9703	9704	CA-2018-147032	31/07/2018	4/8/2018	Standard Class	LB-16795	Laurel Beltran	Home Office	United States	Wilmington	Delaware	19805.0	East	OFF-BI-10003656	Office Supplies	Binders
9704	9705	CA-2018-147032	31/07/2018	4/8/2018	Standard Class	LB-16795	Laurel Beltran	Home Office	United States	Wilmington	Delaware	19805.0	East	OFF-BI-10002897	Office Supplies	Binders
--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
9795	9796	CA-2017-125920	21/05/2017	28/05/2017	Standard Class	SH-19975	Sally Hughesby	Corporate	United States	Chicago	Illinois	60610.0	Central	OFF-BI-10003429	Office Supplies	Binders
9796	9797	CA-2016-128608	12/1/2016	17/01/2016	Standard Class	CS-12490	Cindy Schnelling	Corporate	United States	Toledo	Ohio	43615.0	East	OFF-AR-10001374	Office Supplies	Art
9797	9798	CA-2016-128608	12/1/2016	17/01/2016	Standard Class	CS-12490	Cindy Schnelling	Corporate	United States	Toledo	Ohio	43615.0	East	TEC-PH-10004977	Technology	Phones
9798	9799	CA-2016-128608	12/1/2016	17/01/2016	Standard Class	CS-12490	Cindy Schnelling	Corporate	United States	Toledo	Ohio	43615.0	East	TEC-PH-10000912	technology	Phones

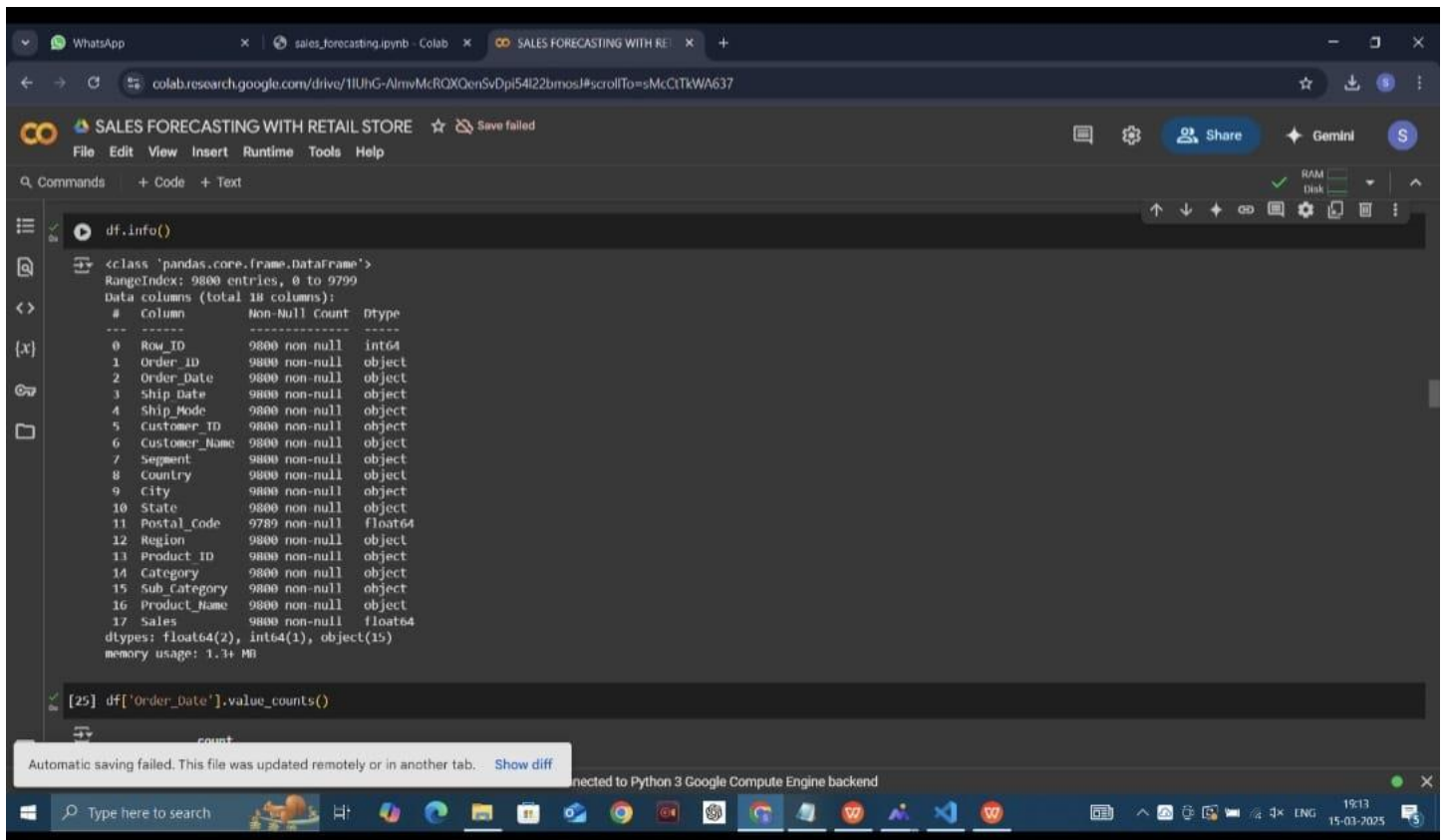
Spaces: 4Cell 9 of 37

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```
print(df.isnull().sum())
```

```
..
Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
Customer_ID 0
Customer_Name 0
Segment     0
Country     0
City        0
State       0
Postal_Code 11
Region      0
Product_ID  0
Category    0
Sub_Category 0
Product_Name 0
Sales       0
dtype: int64
```

df.info()



df['Sales'].value_counts()

WhatsApp

sales_forecasting.ipynb - Colab

SALES FORECASTING WITH RETAIL STORE

+

colab.research.google.com/drive/1Uhg-ALmvMcRQXQen5vDpi54l22bmosJ#scrollTo=jtnMEW4tCmjN

SALES FORECASTING WITH RETAIL STORE

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count

Sales

12.960	55
15.552	39
19.440	39
10.368	35
25.920	34
...	...
339.136	1
60.048	1
5.022	1
7.857	1
10.384	1

5757 rows x 1 columns

dtype: int64

```
df['Order_Date']=pd.to_datetime(df['Order_Date'],errors='coerce')
df['Order_Date']=df['Order_Date'].fillna(method='ffill')
```

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with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.

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1929 15-03-2025

```
[ ] df['Order_Date']=pd.to_datetime(df['Order_Date'],errors='coerce')
df['Order_Date']=df['Order_Date'].fillna(method='ffill')
```

```
<ipython-input-27-d0f10da09b12>:2: FutureWarning: Series.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.
df['Order_Date']=df['Order_Date'].fillna(method='ffill')
```

```
[ ] from matplotlib import pyplot as plt
```

```
[ ] df['year']=df['Order_Date'].dt.year
df['month']=df['Order_Date'].dt.month
df['day']=df['Order_Date'].dt.day_name()
```

```
[ ] df
```

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Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub Category	
9700	9701	CA-2018-154116	15/12/2018	19/12/2018	Standard Class	KM-16660	Khloe Miller	Consumer	United States	Inglewood	California	90301.0	West	TFC-PH-10000675	Technology	Phones
9701	9702	CA-2017-105291	30/10/2017	4/11/2017	Standard Class	SP-20920	Susan Pistek	Consumer	United States	San Luis Obispo	California	93405.0	West	OFF-FA-10003059	Office Supplies	Fasteners
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9704	9705	CA-2018-147032	31/07/2018	4/8/2018	Standard Class	LB-16795	Laurel Beltran	Home Office	United States	Wilmington	Delaware	19805.0	East	OFF-BI-10002897	Office Supplies	Binders
9795	9796	CA-2017-125920	21/05/2017	28/05/2017	Standard Class	SH-19975	Sally Hughsby	Corporate	United States	Chicago	Illinois	60610.0	Central	OFF-BI-10003429	Office Supplies	Binders
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9797	9798	CA-2016-128608	12/1/2016	17/01/2016	Standard Class	CS-12490	Cindy Schnelling	Corporate	United States	Toledo	Ohio	43615.0	East	TFC-PH-10004977	Technology	Phones
9798	9799	CA-2016-128608	12/1/2016	17/01/2016	Standard Class	CS-12490	Cindy Schnelling	Corporate	United States	Toledo	Ohio	43615.0	East	TFC-PH-10000912	Technology	Phones

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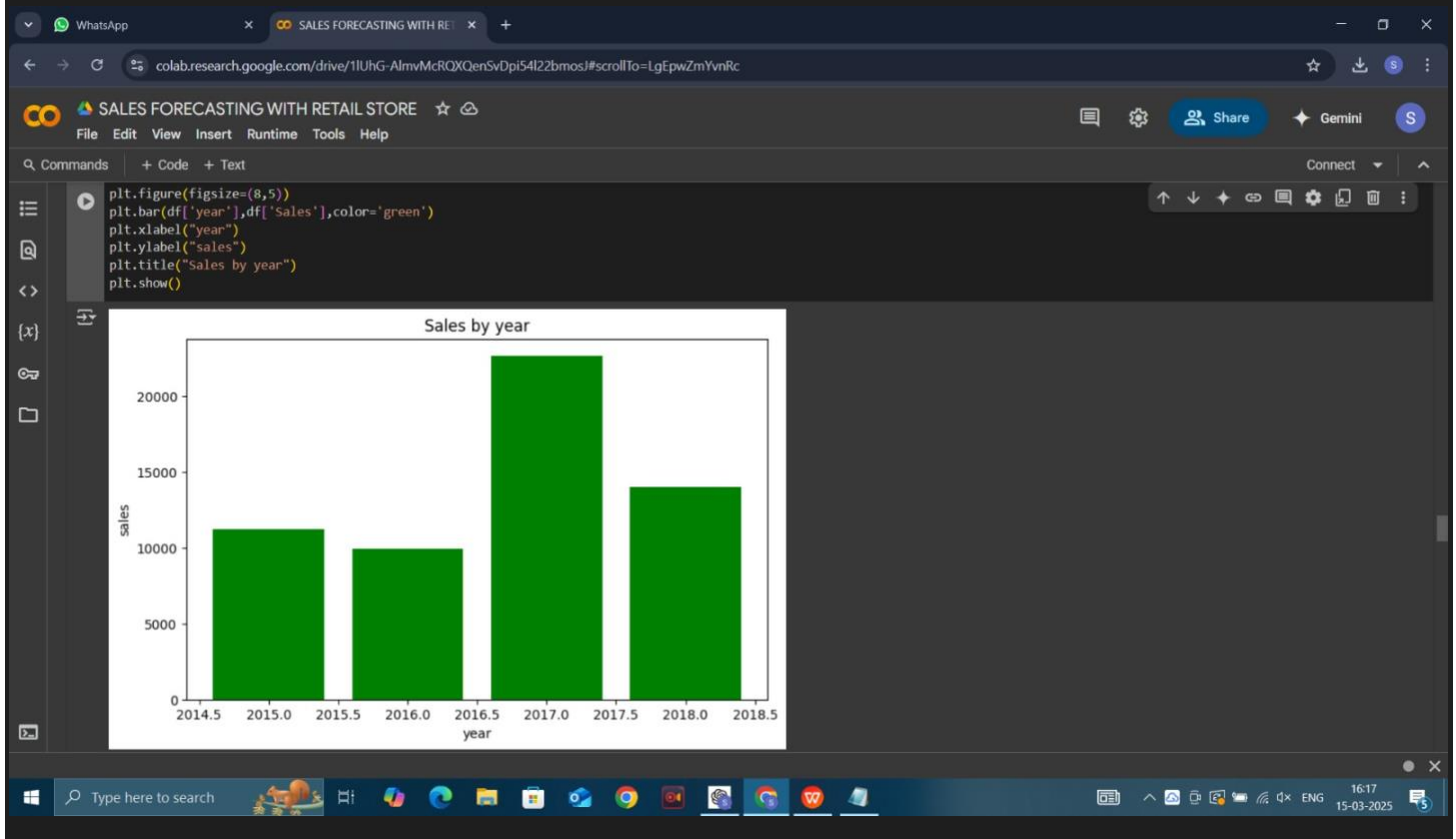
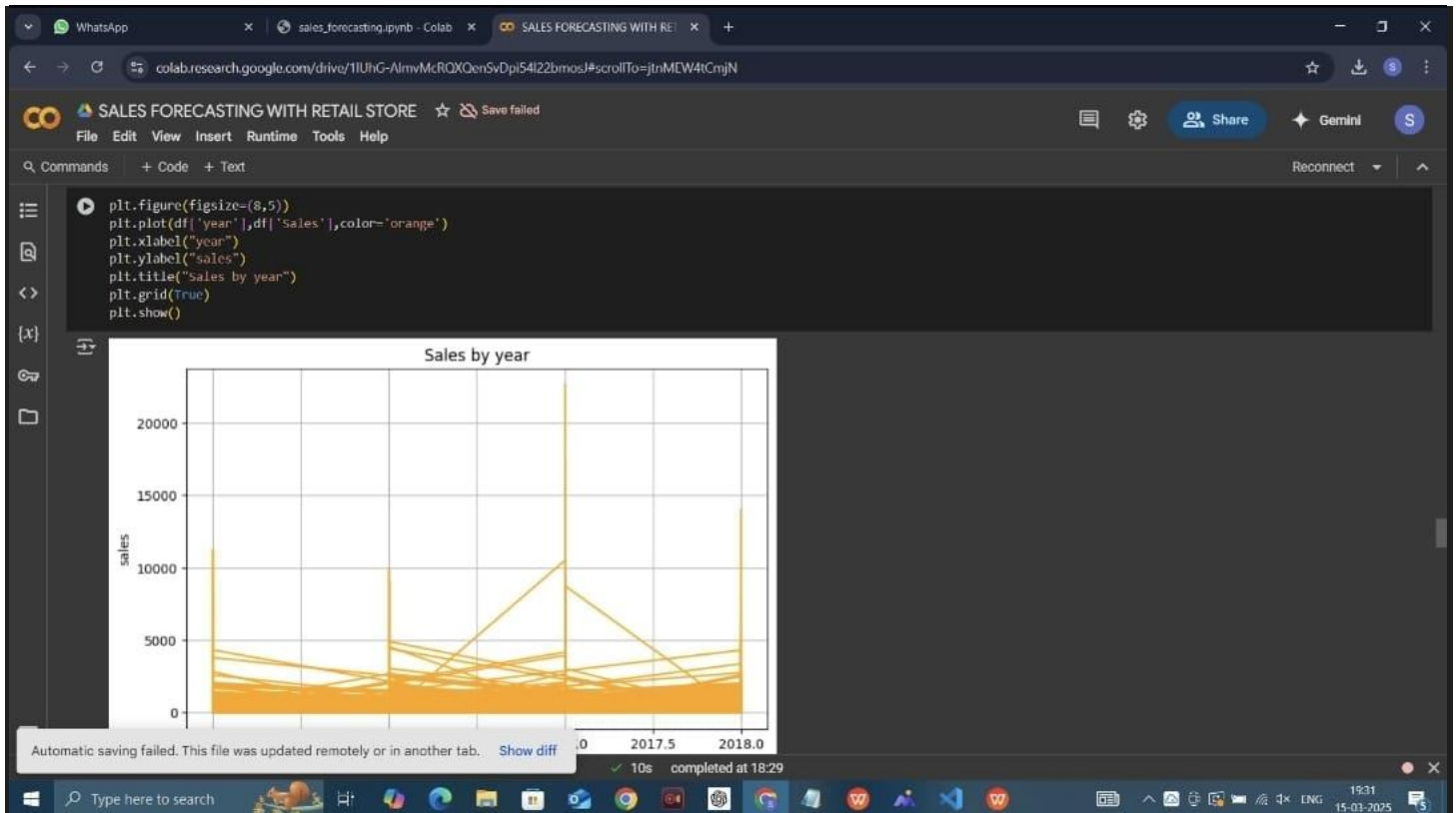
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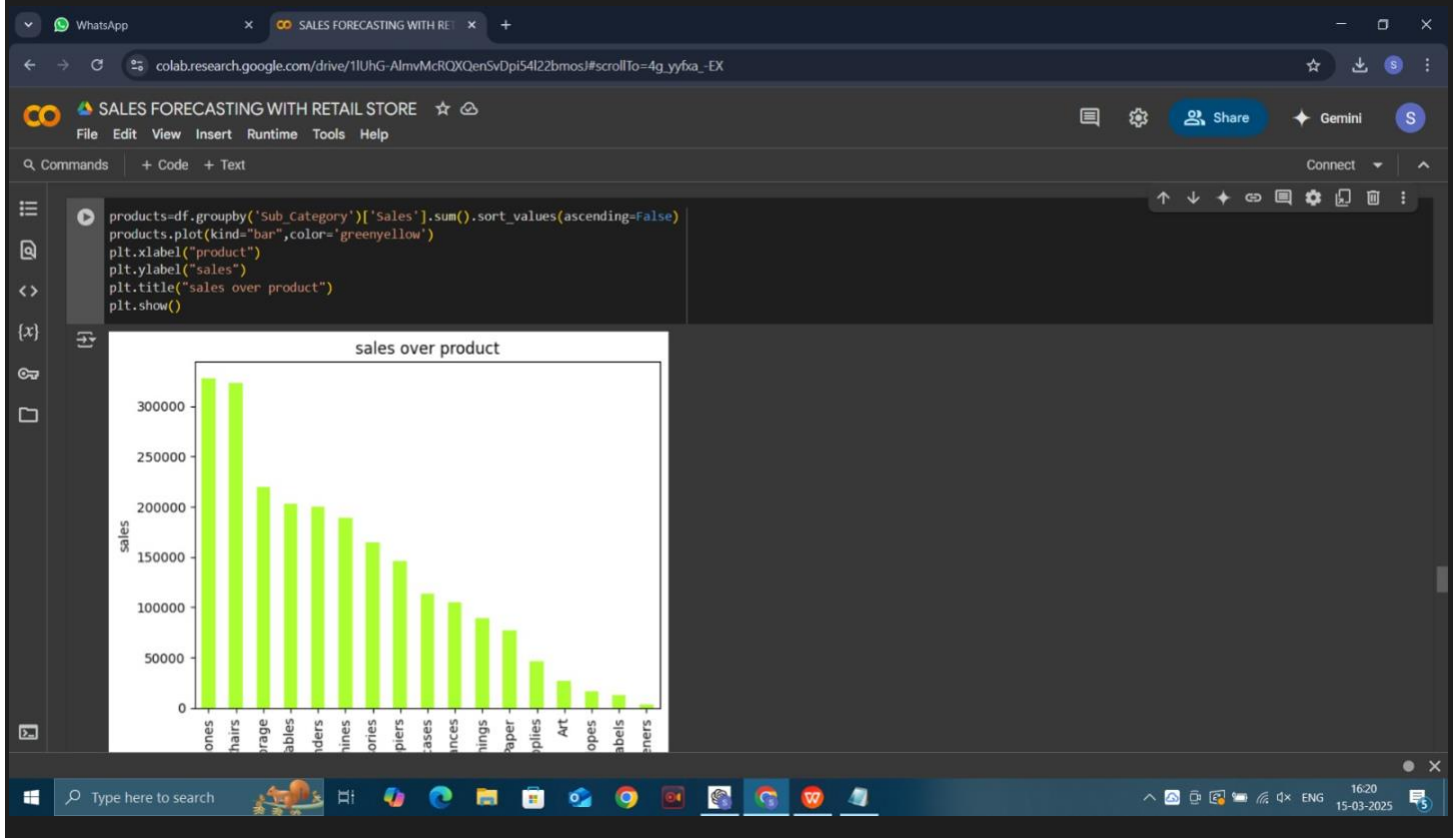
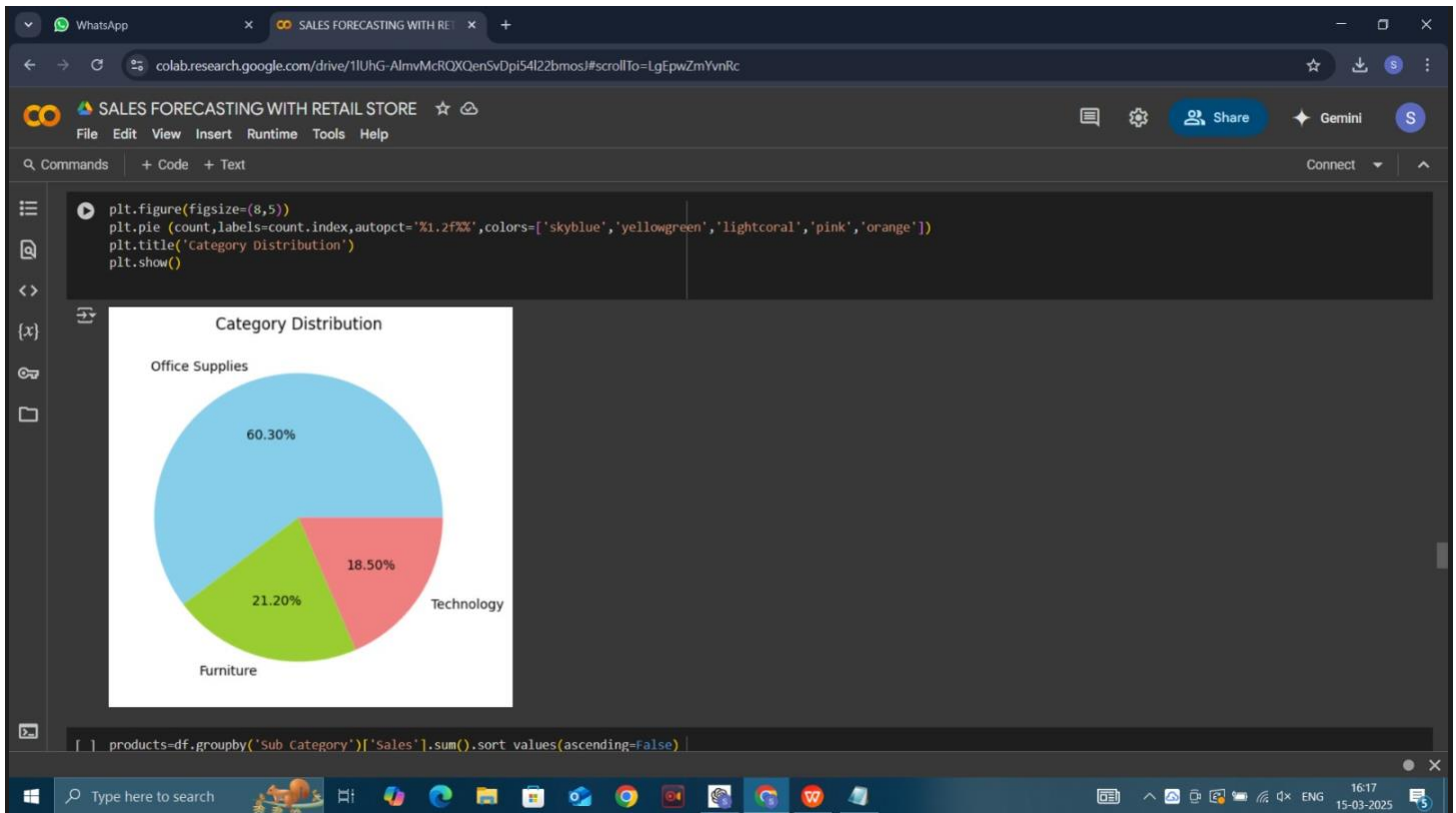
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```
plt.figure(figsize=(8,5))
plt.scatter(x=df['year'],y=df['Sales'],color='blue')
plt.xlabel("year")
plt.ylabel("sales")
plt.title("Sales by year")
plt.grid(True)
plt.show()
```

Sales by year

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```
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[ ] from sklearn.model_selection import train_test_split
    from sklearn.linear_model import LinearRegression

[ ] X= df[['year','month','day']]
    Y= df['Sales']

[ ] print(X.dtypes)
    print(Y.dtypes)

year      int32
month     int32
day       object
dtype: object
float64

print(X.select_dtypes(include=[ 'object' ] ).head())

      day
0  Friday
1  Friday
2 Wednesday
3  Thursday
4  Thursday

[ ] X = pd.get_dummies(X, drop_first=True)

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```
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X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)

+ LinearRegression ⓘ
LinearRegression()

[ ] from sklearn.metrics import mean_absolute_error, mean_squared_error

[ ] y_pred = model.predict(X_test)

[ ] import numpy as np

[ ] MAE= mean_absolute_error(y_test,y_pred)
    MSE= mean_squared_error(y_test,y_pred)
    RMSE= np.sqrt(MSE)

[ ] print(f'Mean Absolute Error : {MAE}')
    print(f'Mean squared Error : {MSE}')
    print(f'Root Mean squared Error : {RMSE}')
```

```
y_pred = model.predict(X_test)

plt.figure(figsize=(10, 5))

plt.plot(y_test.values, label="Actual Sales", marker='o', linestyle='dashed', color='blue')

plt.plot(y_pred, label="Predicted Sales", marker='s', linestyle='solid', color='red')
```

```
plt.xlabel("Data")  
plt.ylabel("Sales")  
plt.title("Actual vs Predicted Sales")  
plt.legend()  
plt.show()
```

